

PORTFOLIO

COOPERATIVE EDUCATION
INTERNSHIP



RATTANUN DEEWONGSAI

รฐนันท์ ตีวงศ์สาย

CHIANG MAI UNIVERSITY - COMPUTER SCIENCE

Profile



Personal Information

Name	Rattanun Deewongsai (รัททันต์ ดีวงศ์สาย)
Nickname	Nortor (นอตอ)
Birth Date	02 October 2004
Age	21
Nationality	Thai
Major	computer science
University	Chiang Mai University
GPA	3.47

"I am a passionate developer focused on building efficient and scalable systems, combining logical problem-solving with creative thinking to deliver impactful digital solutions."

Programming Languages

- Python
- Java
- C / C++
- Go (Golang)
- Dart

Web Technologies

- HTML, CSS
- JavaScript

Strengths

- Problem Solving
- Algorithmic Thinking
- Backend Development Fundamentals
- Collaboration
- Communication Skills
- Creativity

Education

Chiang Mai University

Faculty of Science (2023 - present)
B.S. in Computer Science

Languages

- English (Upper Intermediate)
- Chinese (Intermediate)
- Thai (Native)

Contacts



0841179098



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github.com/NewtronNK



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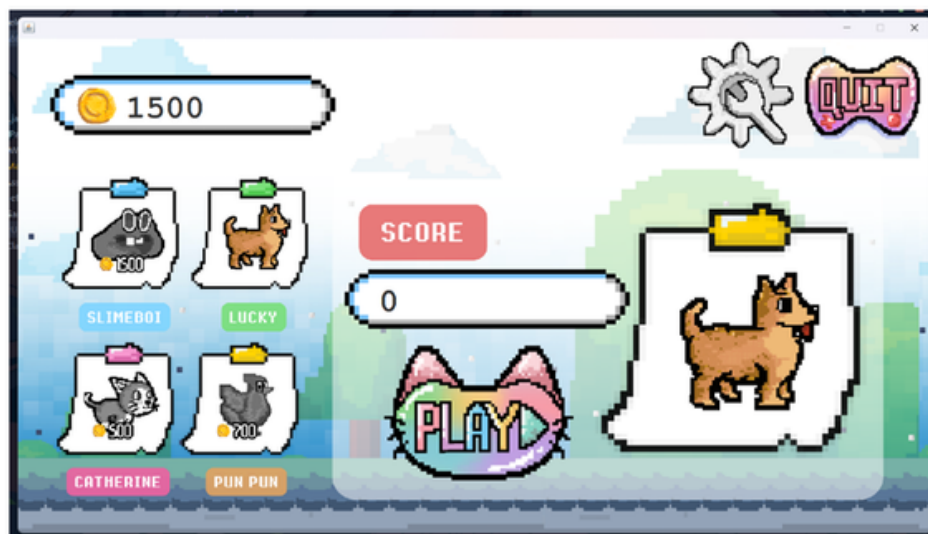
- 01 PROJECTS AND RESEARCH
 - 02 GRAPHIC WORKS
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01 Projects

Petty Run | OOP Game Project (December 2023)

Developed a platformer game using Object-Oriented Programming (OOP) principles in Java as part of a 4-member team.

- Implemented core gameplay mechanics and character logic
- Designed map patterns and level structure
- Contributed to UX/UI implementation
- Awarded 4th place in Popular Vote among 20 competing teams



01 Projects

Netflix Data Cleaning | Big Data Processing Project (February 2024)

This project focused on cleaning and preprocessing a Netflix dataset obtained from Kaggle. The original dataset contained missing and inconsistent values, which required systematic data cleaning before meaningful analysis could be performed.

- Imported and analyzed the Netflix dataset from Kaggle using Python in Jupyter Notebook
- Identified and handled missing values in key attributes such as director, cast, and country
- Performed data transformation and formatting to standardize the dataset
- Prepared structured data suitable for further analysis



netflix_titles.csv (3.4 MB)

Detail Compact Column 10 of 12 columns ▾

About this file

All TV Shows and Movies meta data on Netflix. Updated every month.

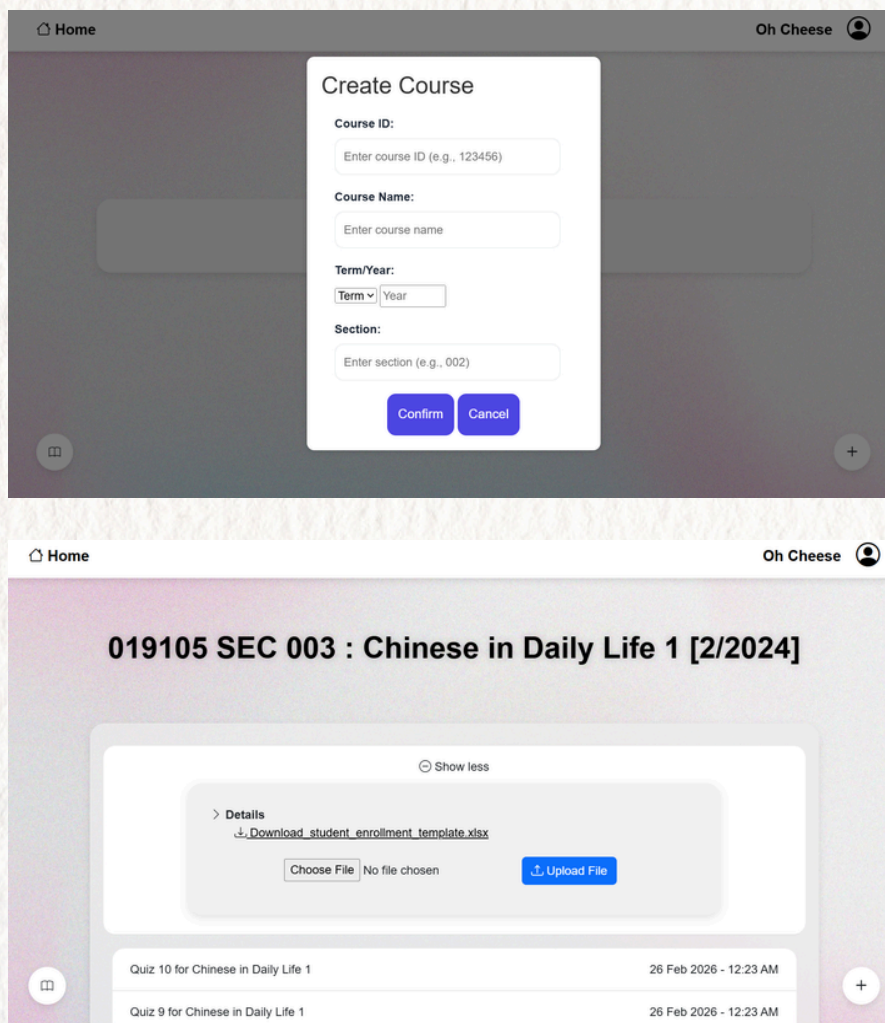
show_id	type	title	director
Unique ID for every Movie / Tv Show	Identifier - A Movie or TV Show	Title of the Movie / Tv Show	Director of the
8807 unique values	Movie TV Show	70% 30%	[null] Rajiv Chilaka Other (6154)

01 Projects

Score Announcement Website | Web Application Development Project (March 2025)

Developed a web application for academic score announcements in a team of 4 members.

- Designed and structured the database system
- Integrated CMU OAuth and Google Authentication for secure login
- Implemented Excel (.xlsx) processing for score table generation
- Built the score announcement module, including file reading, calculation logic, and UI components



Due to university data privacy policies, real user data cannot be publicly disclosed. Therefore, the system demonstration is provided using an admin account with simulated data.

01 Projects

Travel Planner System | Algorithm-Based Project (March 2025)

Developed a travel planning system designed to optimize travel routes based on the number of available days and travel distance constraints.

- Implemented route optimization logic based on distance calculation
- Designed data structures to manage location datasets
- Applied algorithmic problem-solving to maximize travel efficiency

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TravelPlanner.ipynb ☆ ☁
ไฟล์ แก้ไข มุมมอง แท็บ จินตภาพ เครื่องมือ ความช่วยเหลือ
+ โด็ค + ข้อความ ▶ เรียกใช้ทุกเซลล์ ▼

[11] 0
✓ 1 บรรทัด

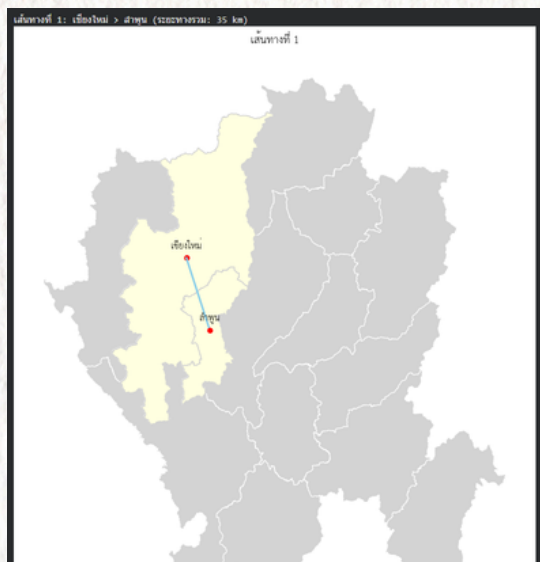
results = travel_planner_astar(locations, start_location, max_day, max_distance, end_location)

# Filter out duplicate paths
filtered_results = []
seen_paths = set()
for path, distance in results:
    # Convert the path to a tuple to make it hashable for the set
    path_tuple = tuple(path)
    if path_tuple not in seen_paths:
        seen_paths.add(path_tuple)
        filtered_results.append((path, distance))

# Now use filtered_results instead of results
if filtered_results:
    print("เริ่มเดินทางจากจังหวัด: " + start_location + \
          "\nจำนวนวันสูงสุด: " + str(int(max_day)) + \
          "\nจำนวนระยะทางสูงสุด(km): " + str(max_distance) + "\n")
    for i, (path, distance) in enumerate(filtered_results, 1):
        print(f"เส้นทางที่ {i}: {' > '.join(path)} (ระยะทางรวม: {distance} km)")

▼ ... เริ่มเดินทางจากจังหวัด: เชียงใหม่
จำนวนวันสูงสุด: 7
จำนวนระยะทางสูงสุด(km): 1000000

เส้นทางที่ 1: เชียงใหม่ > ลำพูน (ระยะทางรวม: 35 km)
เส้นทางที่ 2: เชียงใหม่ > ลำปาง > ลำพูน (ระยะทางรวม: 306 km)
เส้นทางที่ 3: เชียงใหม่ > เชียงราย > ลำปาง > ลำพูน (ระยะทางรวม: 501 km)
เส้นทางที่ 4: เชียงใหม่ > เชียงราย > พะเยา > ลำปาง > ลำพูน (ระยะทางรวม: 517 km)
เส้นทางที่ 5: เชียงใหม่ > เชียงราย > พะเยา > น่าน > แพร่ > ลำปาง > ลำพูน (ระยะทางรวม: 907 km)
เส้นทางที่ 6: เชียงใหม่ > เชียงราย > พะเยา > น่าน > อุตรดิตถ์ > แพร่ > ลำปาง > ลำพูน (ระยะทางรวม: 1080 km)
เส้นทางที่ 7: เชียงใหม่ > แม่ฮ่องสอน > ตาก > สุโขทัย > ลำปาง > ลำพูน (ระยะทางรวม: 1144 km)
เส้นทางที่ 8: เชียงใหม่ > เชียงราย > พะเยา > น่าน > อุตรดิตถ์ > สุโขทัย > ลำปาง > ลำพูน (ระยะทางรวม: 1171 km)
เส้นทางที่ 9: เชียงใหม่ > แม่ฮ่องสอน > ตาก > สุโขทัย > อุตรดิตถ์ > แพร่ > ลำปาง > ลำพูน (ระยะทางรวม: 1213 km)
เส้นทางที่ 10: เชียงใหม่ > แม่ฮ่องสอน > ตาก > กำแพงเพชร > สุโขทัย > ลำปาง > ลำพูน (ระยะทางรวม: 1353 km)
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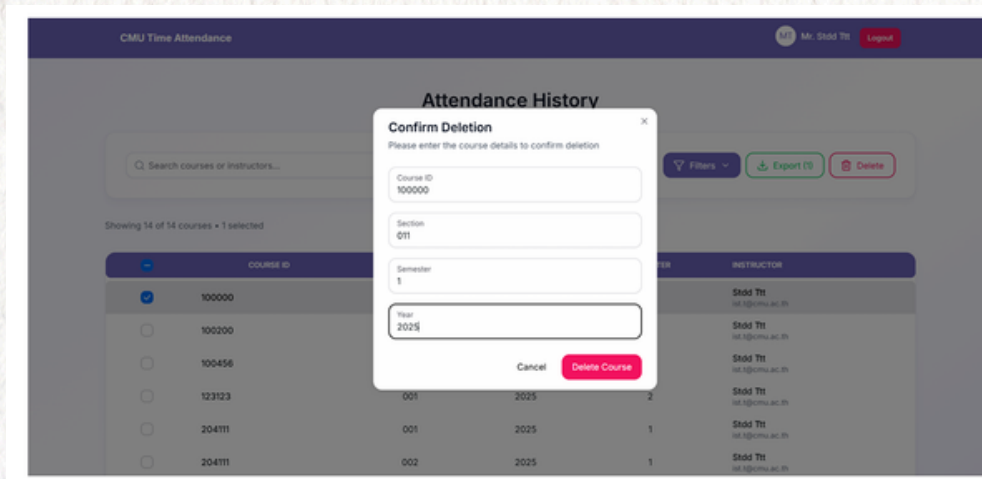
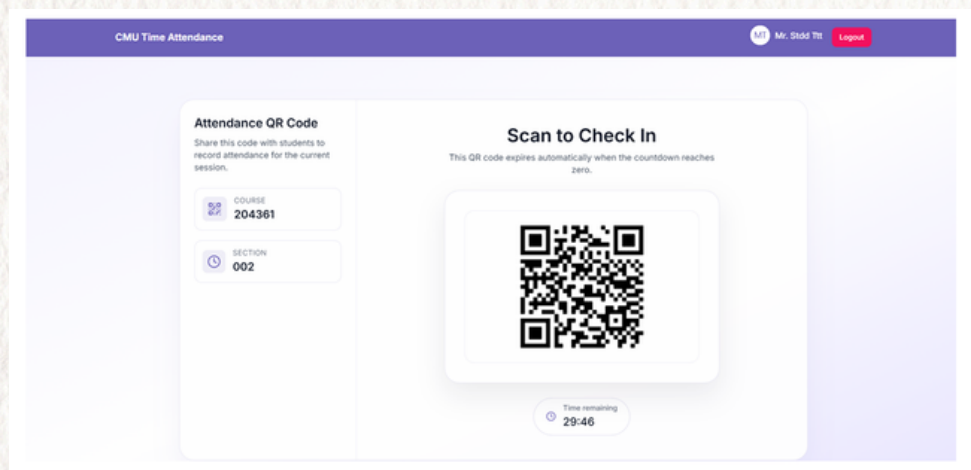
01 Projects

Attendance Checking Website | Software Engineering Project (October 2025)

Developed a classroom attendance web application for a real client in a 6-member team.

- Implemented two-step verification (authentication + location validation)
- Designed and managed the database system
- Developed Excel (.xlsx) integration for student roster and score management
- Conducted data mapping for customer requirements

Project received a 9/10 customer satisfaction rating, demonstrating reliability and practical usability.

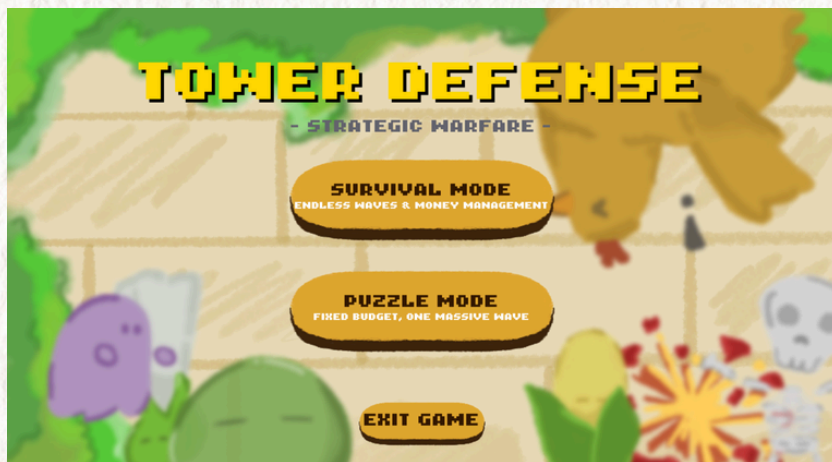


01 Projects

Tower Defense Game | Algorithm-Based Game Project (March 2026)

Developed a tower defense game featuring dynamic pathfinding using Flow Field Navigation combined with graph theory algorithms.

- Implemented Breadth-First Search (BFS) to calculate the shortest path for standard enemies.
- Applied Dijkstra's Algorithm for smart enemies to avoid tower attack zones by considering movement cost.
- Modeled enemy movement as a graph pathfinding problem and optimized routes based on environmental constraints.
- Implemented a feasibility check system to prevent tower placements that would completely block all possible paths to the base.

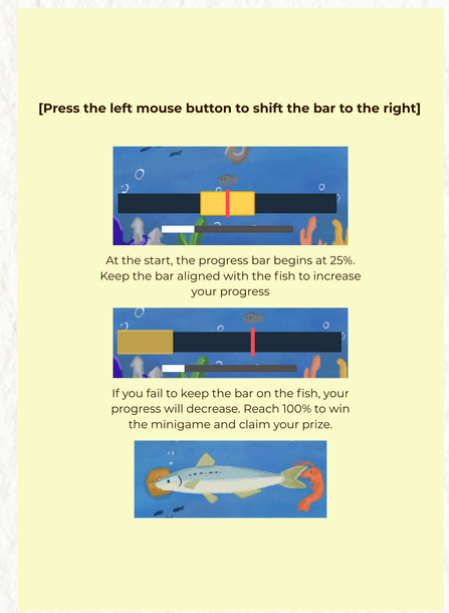
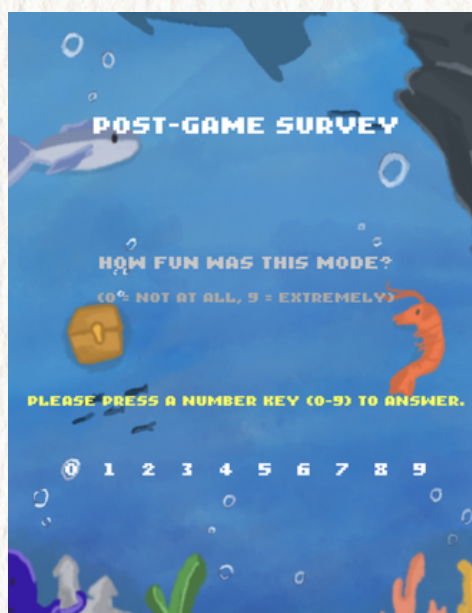
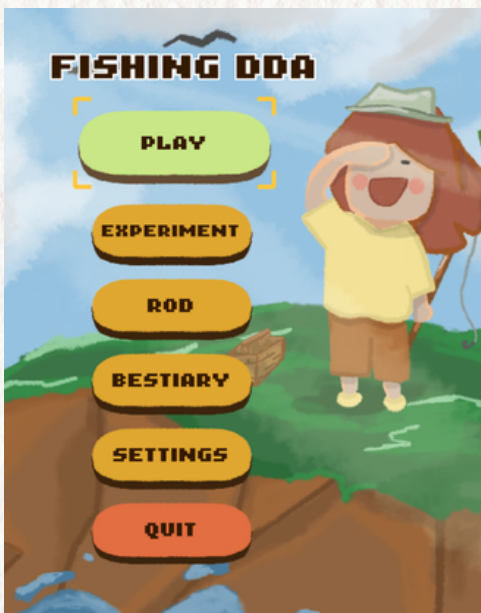


01 RESEARCH

DDA (Dynamic Difficulty Adjustment) | Research Project (March 2026 – Present)

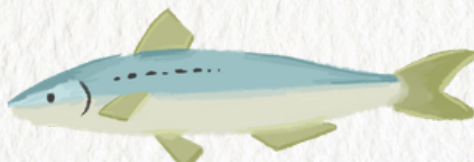
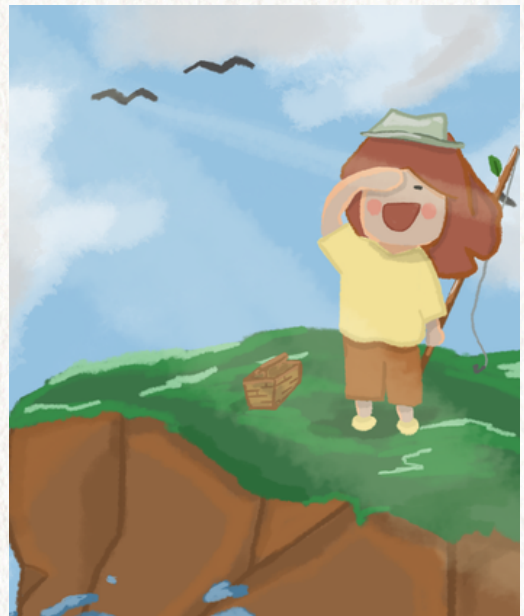
Conducting research on Dynamic Difficulty Adjustment (DDA) through a fishing game simulation to analyze player experience and engagement under varying difficulty levels and reward systems.

- Designed and implemented a simulation model to dynamically adjust game difficulty
- Developed logic for adaptive reward mechanisms based on player performance
- Collected and analyzed gameplay data to evaluate user engagement and satisfaction
- Applied experimental design principles to measure player response across different difficulty configurations



02 GRAPHICS

Designed visual assets and artwork for the Dynamic Difficulty Adjustment (DDA) research project, including UI elements, game assets, and presentation materials.



02 GRAPHICS

Designed the official annual class shirt for the Computer Science Department.



02 GRAPHICS

Designed the Advisee 66X banner for the Computer Science Department.



Thank You
Thank you for your
time and
consideration.

I look forward to the
opportunity to contribute to
your organization.